

Activity - 2

① Medical Image Enhancement

Questions

1) Which gray - - -

Ans Power-law (Gamma) Transformation.

2) Why is it suitable?

Ans It enhances dark regions without excessively affecting bright regions, making soft tissues more visible in the x-ray.

3) What effect - - -

Ans Dark tissue become brighter & clearer while bones remain visible, improving the overall image quality.

② Night-Time Surveillance

Questions

① Which - - -

Ans log transformation

② Why is it suitable.

Ans log transformation expands low-intensity (dark) pixel values, making dark areas brighter. Negative transformation only inverts the image & does not improve visibility.

③ Predict the - - -

Ans The Parking lot will appear brighter, and objects such as vehicles, people, and surroundings will be easier to identify.

③ Finger Print Recognition

1) Which gray - - -

Ans Contrast stretching (Piecewise-Linear Transformation)

Q4 Explain why?

A4 It increases the contrast between fingerprint ridges and valleys, making the fine ridge details more distinct for accurate recognition.

Q5 What happens

A5 log transformation would brighten the darker regions but would not effectively enhance the ridge-valley contrast. As a result, fingerprint details may become less distinct compared to Piecewise Linear Transformation.